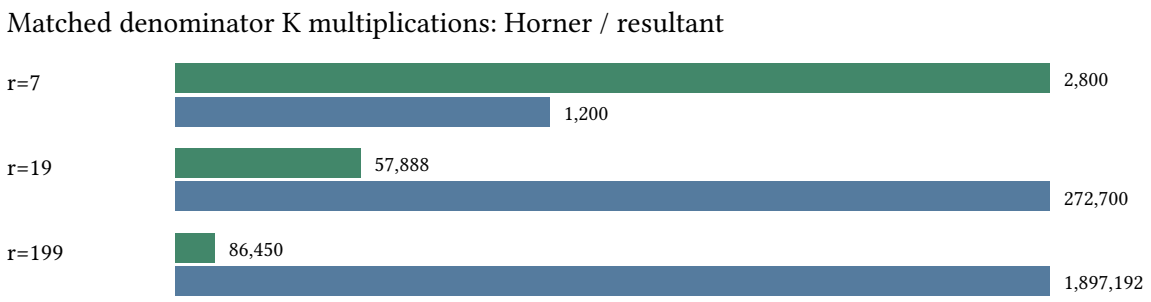
Fourth-power denominator norm



Source: denominators.json. Separate row scales; green Horner, blue resultant. No timing claim.

Derived and qualified 2026-11-09, T-1968. The signed baby denominator norm can be evaluated by two polynomial evaluations and fourth powers, with no field inverse. The contract is qualified independently of the complete first-response pipeline. All scientific results remain unverified producer evidence. [pcc:unverified]

Proof. Let H_G(Z)=product_j(Z-x(jG)), with distinct baby roots, and beta(x(jG))=x(jQ). Define H_Q(Z)=product_j(Z-x(jQ)). For giant inputs A,B put D=(x(A)-Z)^2*(x(B)-beta(Z))^2 in K[Z]/H_G. Evaluating D^2 at every root gives Norm(D^2)=product_j(x(A)-x(jG))^4*(x(B)-x(jQ))^4=H_G(x(A))^4*H_Q(x(B))^4. The nonrational-shift guards from Round63 ensure every factor is nonzero. This is standard norm multiplicativity applied to the retained signed input; it does not require constructing D^2 or computing its resultant. [pcc:unverified]

Frozen panel. All 11,382 full baby blocks from the unchanged Round63/64 query inputs match the field-resultant denominator value. Polynomial preparation, target H_Q setup and each isolated evaluator have separate counters. The evaluation path records 147,138 K multiplications and zero inversions, versus 2,171,092 multiplications and 67,255 inversions in the resultant path. Target H_Q construction costs 496 additional multiplications. The degree-one panel uses 2,800 evaluation multiplications versus 1,200 in the reference; all panels remain in the record. No timings or complete-evaluator speedup are inferred. [pcc:unverified]

Primary P192 checks. The 72 prescribed degree-four denominator norms on the existing publicQ,mu_G,relation_lambda_G all match. The two paths use 1,080/0 and 2,808/360 explicit K multiplication/inverse calls respectively. These counters exclude inverse internals and are not combined with the small-field model. Polynomial inputs and curve points are explicitly prepared; no full P192 response or arbitrary-input decoder is executed. [pcc:unverified]

Custody. Scope and source precede the data. Pre-data review corrected nested quadratic-point parsing and added whole-qualifier counters; both source commits remain. Fresh-copy outputs and a changed-record rejection are retained in validation.json. The unchanged legacy library timeout is inherited from Round63 rather than retried. No cloud experiment, generator or old freeze lane was started or modified. [pcc:unverified]

Next integration. Cache H_Q per changing target and use this denominator evaluator in the selected norm response, with its constructor and any memory added to the ledger. Numerator formation and numerator contraction across giant blocks remain the larger compression question. The previous norm and decoder outputs are unchanged historical evidence; this qualification does not substitute its isolated count for total response cost. [pcc:unverified]

Sources checked. Round63 paired-algebra-proof.md and Round64 resultant-proof.md supply the concrete denominator and guards. Norm-as-product background is in Stacks0BIE, https://stacks.math.columbia.edu/tag/0BIE . KB tools/wrapper are unavailable. No novelty/absence claim for norm multiplicativity or Horner evaluation. [pcc:unverified]